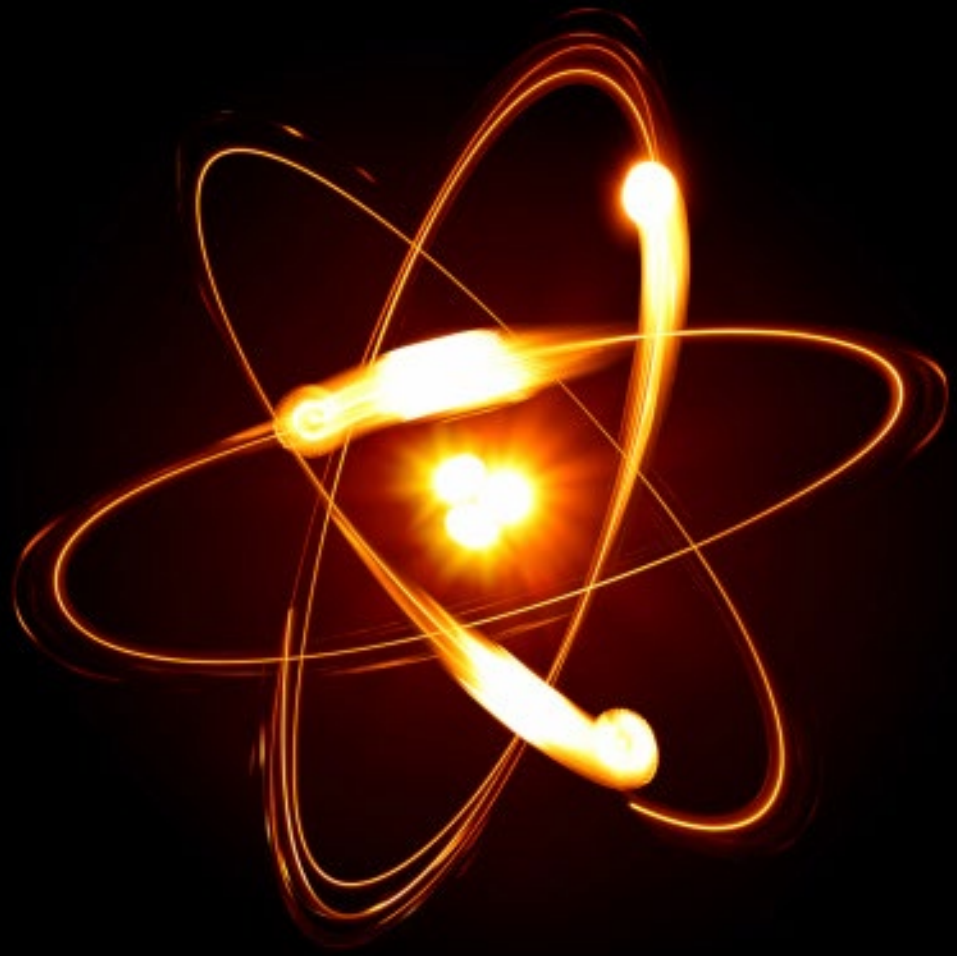
Chapter 25.

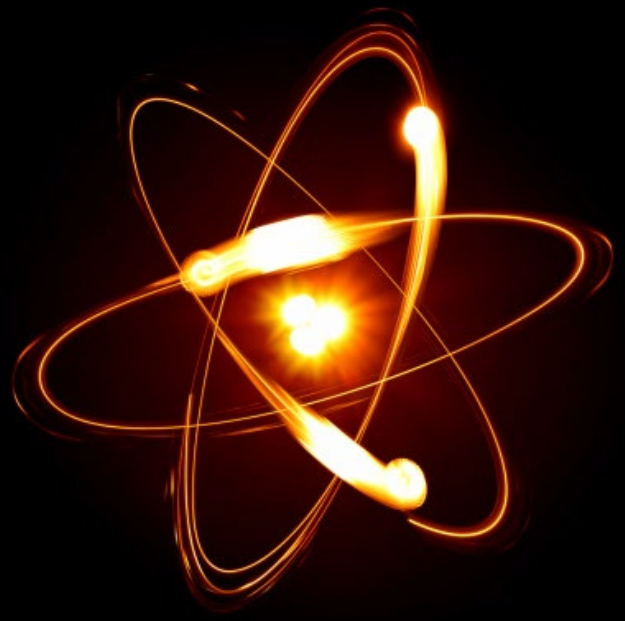
Many-electron Atoms

Part 2



Beatriz González del Río

Quantum Physics



By convention sweet is sweet, bitter is bitter, hot is hot, color is color; but in truth there are only atoms and the void.

Democritus of Abdera
(c 400 BC)

Chapter 25.

Many-electron Atoms. Part 2

- Fine Structure and Spin–Orbit Interaction
- Optical Excitations in Many-Electron Atoms
- Alkali Atoms as a Model System
- Atoms with Multiple Optically Active Electrons
- LS Coupling and Term Structure
- Optical Transitions and Selection Rules

Bibliography

Eisberg and Resnick; B H Bransden and C J Joachain, Physics of Atoms and Molecules, Prentice Hall, 2003 ; Alonso and Finn (Vol. 3) ; H A Bethe and R Jackiw, Intermediate Quantum Mechanics, Addison Wesley, 1997

Magnetics effects and fine structure

With more detailed resolution, the spectral lines of **hydrogenic (alkali, too) atoms** are in fact doublets. Only s–states are not split. Interpretation?

Magnetic effects!!

Connected to the orbital angular momentum, we have an orbital magnetic dipole moment

$$\vec{\mu}_l = -\frac{e}{2m_e}\vec{L} = -g_l\mu_B\frac{\vec{L}}{\hbar}$$

where $\mu_B = \frac{e\hbar}{2m_e}$ (Bohr's magneton) and $g_l = 1$.

Also, associated to the spin of the electron, there is a spin magnetic dipole moment $\vec{\mu}_s$.

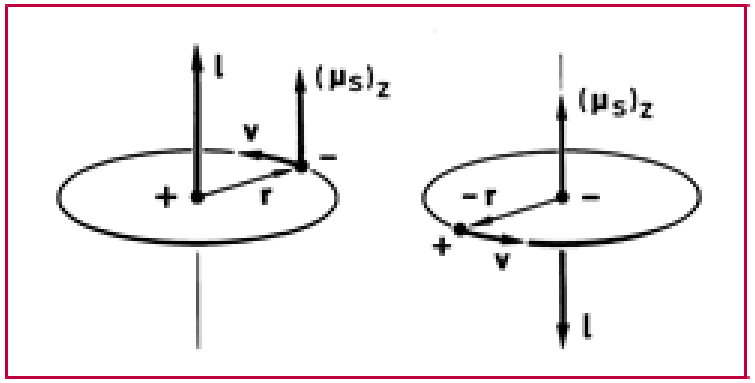
$$\vec{\mu}_s = -g_s\mu_B\frac{\vec{S}}{\hbar} \qquad g_s = 2$$

We may consider that the nucleus revolves around the electron. This generates a magnetic field $\vec{B}_l$ at the position of the electron. But the electron has a spin magnetic dipole moment $\vec{\mu}_s$ which can take different orientations with respect to $\vec{B}_l$ and this implies different potential energy values $W = -\vec{\mu}_s \cdot \vec{B}_l$

Simple derivation of spin-orbit interaction

We have a magnetic field $\vec{B}_l$ at the site of the electron. (Ampere's law).

$$\vec{B}_l = \frac{\mu_0}{4\pi} Ze \frac{\vec{v} \times (-\vec{r})}{r^3} = \frac{\mu_0}{4\pi} \frac{Ze}{m_e r^3} \vec{L}$$

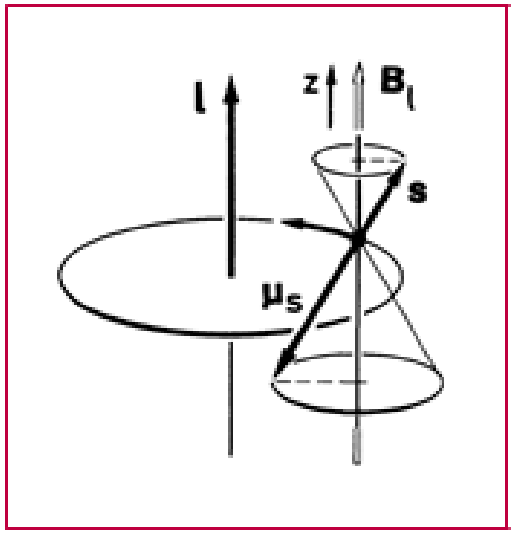


$\vec{L} \Rightarrow$ orbital angular momentum of the electron

$\vec{E} \Rightarrow$ Electric field created by the nucleus $\quad \vec{E} = \frac{Ze}{4\pi\varepsilon_0} \frac{\vec{r}}{r^3}$

Force acting on the electron $\quad \vec{F} = (-e)\vec{E}$

If the electrostatic potential associated with $\vec{F}$ is spherically symmetric (one electron atom) then $\vec{F} = -\frac{dV(r)}{dr} \frac{\vec{r}}{r}$



Therefore $\vec{B}_l = -\varepsilon_0 \mu_0 (\vec{v} \times \vec{E}) = -\frac{1}{c^2} (\vec{v} \times \vec{E})$

$\vec{B}_l$ is parallel to $\vec{L}$ (and therefore, anti-parallel to $\vec{\mu}_l$)

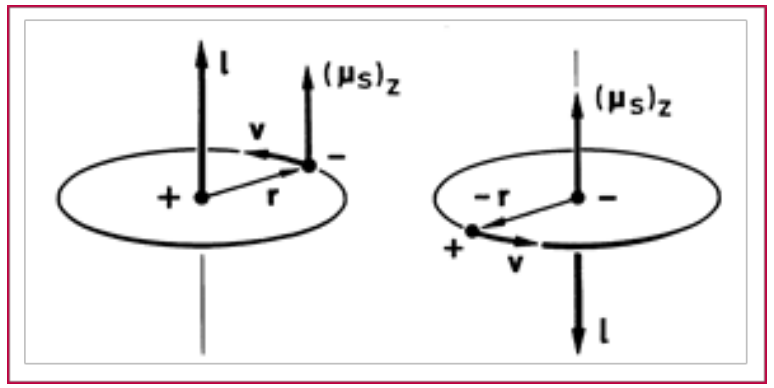
The interaction between $\vec{B}_l$ and $\vec{\mu}_s$ gives rise to a potential energy

$$W_{LS} = -\vec{\mu}_s \cdot \vec{B}_l = \frac{g_s \mu_B}{\hbar} (\vec{S} \cdot \vec{B}_l)$$

An exact relativistic treatment leads to a factor 1/2 so

$$W_{LS} = \frac{1}{2} \frac{g_s \mu_B}{\hbar} (\vec{S} \cdot \vec{B}_l) = \frac{\mu_0}{8\pi} \frac{Ze^2}{m_e^2 r^3} (\vec{L} \cdot \vec{S})$$

Simple derivation of spin-orbit interaction



Taking into account that

$$\vec{B}_l = -\frac{1}{ec^2}\frac{1}{r}\frac{dV(r)}{dr}[\vec{v}\times\vec{r}] = \frac{1}{em_e c^2}\frac{1}{r}\frac{dV(r)}{dr}\vec{L}$$

$$W_{LS} = \frac{g_s\mu_B}{\hbar}(\vec{S}\cdot\vec{B}_l)$$

$$W_{LS} = \frac{1}{2m_e^2c^2}\frac{1}{r}\frac{dV(r)}{dr}(\vec{L}\cdot\vec{S})$$

$$\vec{J} = \vec{L} + \vec{S} \Longrightarrow \vec{L}\cdot\vec{S} = \frac{1}{2}(J^2 - L^2 - S^2)$$

$$W_{LS} = a(\vec{L}\cdot\vec{S}) = \frac{a}{2}[J^2 - L^2 - S^2]$$

So, the shift in the energy is

$$\Delta E = \frac{\langle a\rangle}{2}[j(j+1) - l(l+1) - s(s+1)]\hbar^2$$

$\langle a\rangle$ represents the mean value of **a** evaluated for the electronic state.

The energy of an *L*−state depends also on *J*!

Atomic states with the same *n* and *l* but different *j*-values, have slightly different energies.

Splitting due to spin-orbit interaction

Let's estimate the magnitude of the orientational potential energy for the n=2, l=1 state of the hydrogen atom.

We know that the possible values of *j* are: *l* + 1/2 and *l* – 1/2

We have $V(r) = -\frac{e^2}{4\pi\varepsilon_0}\frac{1}{r}$

Then $W_{LS} = \frac{e^2}{4\pi\varepsilon_0 2m_e^2c^2}\left\langle\frac{1}{r^3}\right\rangle\frac{1}{2}[J^2 - L^2 - S^2]$

Now, we have to evaluate the expectation value of $\frac{1}{r^3}$ for the state n=2, l=1. The result is $\approx \frac{Z^3}{(3a_0)^3}$

We have j = 3/2, 1/2 and the evaluation leads to

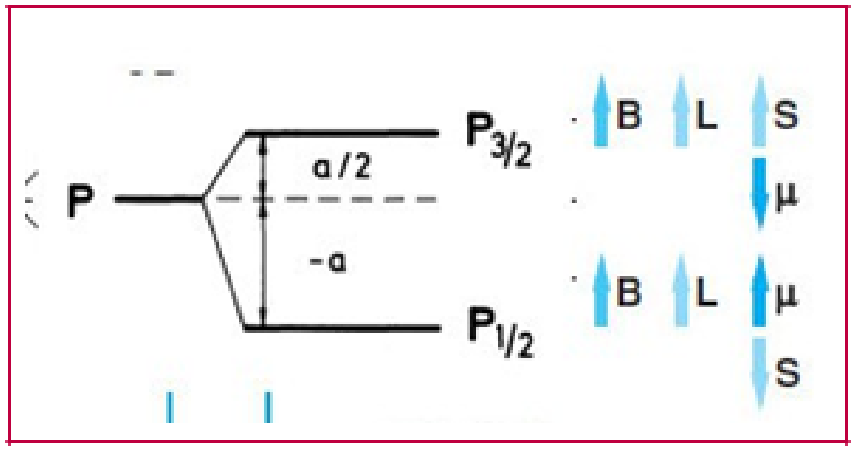
$j = 3/2$ $\quad\quad$ $\Delta E_1 \propto 10^{-23}$ joules

$j = 1/2$ $\quad\quad$ $\Delta E_2 \propto -2\Delta E_1$

So $\Delta E_1 - \Delta E_2 \approx 10^{-23}$ joules $\approx 10^{-4}$ eV. (for comparison $E_{n=2,l=1,m_l} = -3.4$ eV)

Splitting due to spin-orbit interaction

With $Z{\sim}1$ and $r{\sim}1.0$ Å, we obtain that $W_{LS}{\sim}10^{-4}$ eV, which justifies the perturbation treatment for hydrogen and also for the outermost electron of alkali atoms, and $B_l{\sim}1$ Tesla.



The state with higher *J* is less stable, because $\vec{S}$ is "parallel" to $\vec{L}$, and so $\vec{\mu}_s$ is aligned anti-parallel to $\vec{B}_l$. The opposite happens for the lower *J* value.

The *S*–states do not split, because there is no internal field with which the spin can align itself.

The LS splitting increases as Z^4 with nuclear charge and also increases with decreasing *n*. Therefore, it is most important for the innermost electrons.

Splitting due to spin-orbit interaction

EXAMPLE 7-4 Fine-Structure Splitting The fine-structure splitting of the $2P_{3/2}$ and $2P_{1/2}$ levels in hydrogen is 4.5×10^{-5} eV. From this, estimate the magnetic field that the $2p$ electron in hydrogen experiences. Assume $\mathbf{B}$ is parallel to the z axis.

SOLUTION

1. The energy of the $2p$ electrons is shifted in the presence of a magnetic field by an amount given by
$$U = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu_z B$$

2. U is positive or negative depending on the relative orientation of $\boldsymbol{\mu}$ and $\mathbf{B}$, so the total energy difference ΔE between the two levels is:
$$\Delta E = 2U = 2\mu_z B$$

3. Since the magnetic moment of the electron is μ_B, $\mu_z \approx \mu_B$ and
$$\Delta E \approx 2\mu_B B$$

4. Solving this for B substituting for μ_B and the energy splitting ΔE gives
$$B \approx \frac{\Delta E}{2\mu_B} \approx \frac{4.5 \times 10^{-5}\,\text{eV}}{(2)(5.79 \times 10^{-5}\,\text{eV/T})} \approx 0.39\,\text{T}$$

Remarks: This is a substantial magnetic field, nearly 10,000 times Earth's average magnetic field.

Multielectron atoms. Optical excitations

The description of the electrons in a multielectron atom implies the assumption of several approximations:

- Hartree approximation (accounts for the stronger interactions). Electrons moving in an effective spherically symmetric potential

- There are other (weaker) interactions leading to: coupling among the orbital angular momenta, coupling among the spin angular momenta, spin-orbit coupling etc

Taking into account these (weaker) interactions allows to explain several important **properties of the ground state and the low-energy excited states**

Multielectron atoms. Optical excitations

Low-energy excited states?

An atom is raised from ground state to one of its low-energy excited states when an electron in an outer subshell is given a small amount of energy (a few eV)**(optical excitations)** (i.e. a collision between two atoms).
In the deexcitation process, the atom returns to its ground state by emitting some low-energy photons whose frequencies constitute its **optical line spectrum**

The study of the low energy excitations is important for a better understanding of the ground state.

Optical line spectra have practical interest, i.e. Astrophysics.

In the emission spectra of stars, the pattern of lines gives information about its composition. The intensity of lines gives information about the temperature. The Doppler shift informs about its velocity

We begin the study of the optical excitations of multielectron atoms with the simplest case: Alkali atoms

An example: alkali atoms

The alkali atoms have a single weakly bound *s*–electron, while the remaining (Z-1) are in closed subshells. They are in this sense similar to hydrogen.

The highest filled subshell is always a p-type and there is a single s electron in the next s subshell.

Remember that the energy of an electron in a filled *p* subshell is much more negative than the energy of an electron in the next *s* subshell.

Therefore, the *p*-subshell electrons are **not** excited in any of the low energy excitations leading the optical spectra.

Alkalis have a very low ionization potential as compared to the inert gases with one electron less, and this is due to two facts: the single electron occupies a more expanded orbital (a new shell *n* is opened); it is very efficiently shielded from the nuclear attraction by the *core*–electrons (screening).

The energy of the ”single electron” in alkali atoms thus depends on both *n* and *l*.

An example: alkali atoms

- As the energy of the electrons in the highest filled *p*-subshell is much more negative than the energy of the electron in the next *s*-subshell, then the *p*-type electrons are NOT excited by the low-energy excitations leading to the optical spectra.

- Therefore, analysis of the optical spectra of an alkali atom can be made in terms of the outer *s* electron (the other filled subshells can be ignored)

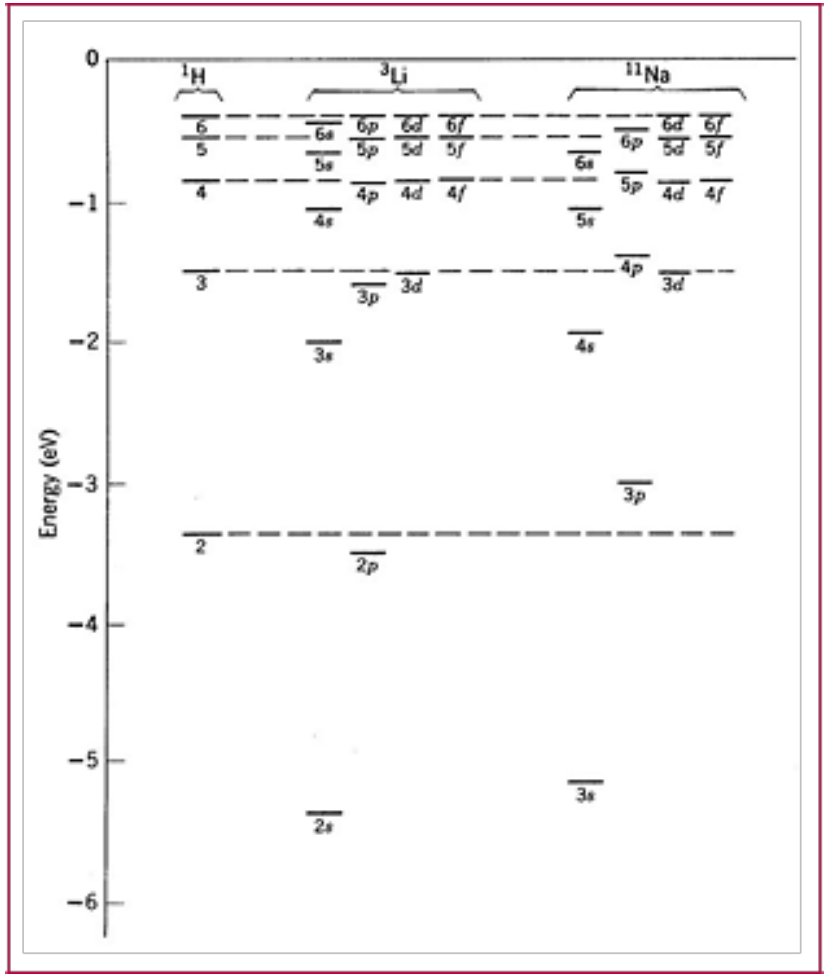
- Basically, an alkali atom may be modelled as an inert core plus a single electron moving in the external subshell.

- The analysis of the optical line spectrum of an alkali atom is very simple because the excited states can be described in terms of the so-called optically active electron and the core of filled subshells can be ignored.

An example: alkali atoms

The total energy of the atom is taken as that of the optically active electron

Each level is labelled by the quantum numbers (n,l) of the optically active electron.



Some energy levels of H, Li and Na atoms

The figure does not include the fine-structure splittings due to the spin-orbit interaction

An example: alkali atoms

- Hartree's model is good at evaluating the energy levels associated to the low-energy excitations. This is so because the net potential due to the nucleus plus the other electrons (those of the filled subshells) is really spherically symmetrical. We only care about the outer *s*-electron.

- The energy of the optically active electron depends on n and l . For a given n, the energy is most negative for the smallest l. However, for Na the l-dependence makes the 4*s* level more negative than the 3*d* level.

- For large n-values, the l dependence becomes less important and the energy levels of the optically active electron become more similar to those of hydrogen. Why? Because the shielding of the nuclear charge Z by the (Z-1) electrons becomes more effective

- The lines of the optical spectra of the alkalis have a fine structure splitting which indicates that all energy levels are **double**, except those for $l = 0$. This is due to the spin-orbit interaction acting on the optically active electron.

An example: alkali atoms, Spin-orbit interaction

- Remember

$$\Delta E = \frac{\langle a \rangle}{2}[j(j+1) - l(l+1) - s(s+1)]$$

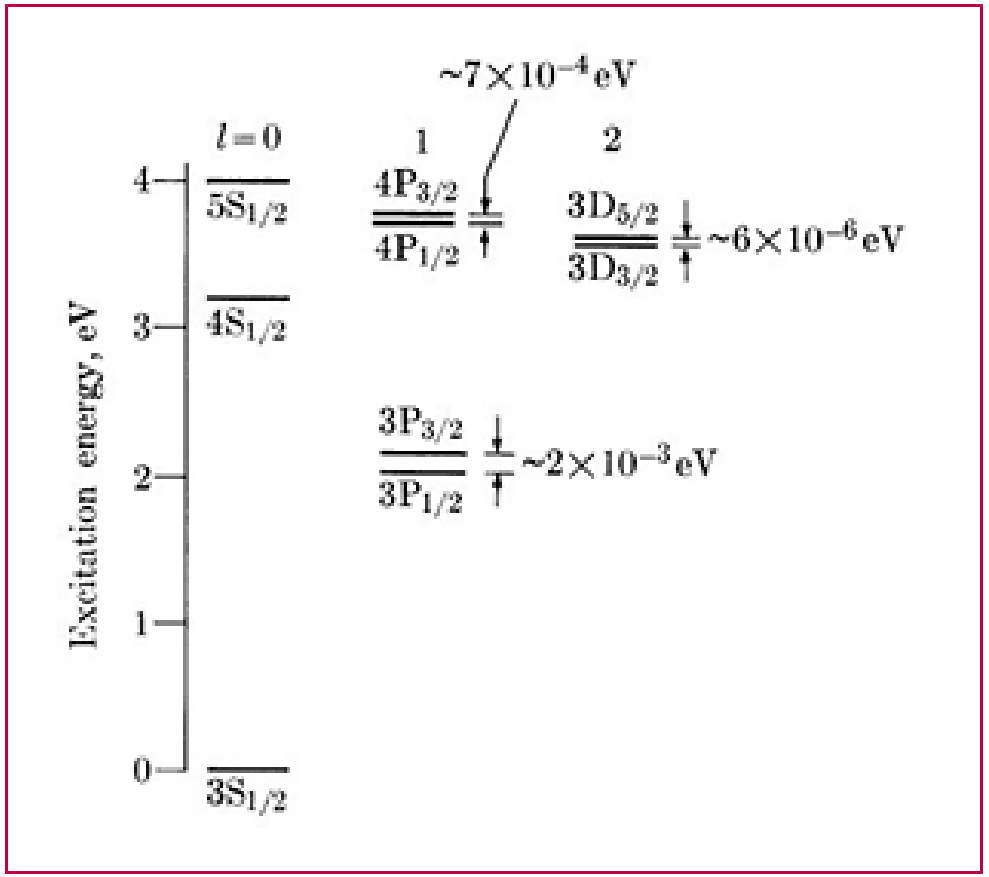
- $$\frac{\langle a \rangle}{2} = \frac{\hbar^2}{4m_e^2 c^2}\left\langle \frac{1}{r}\frac{dV(r)}{dr} \right\rangle$$

where $V(r)$ is identified as the Hartree potential.

- ΔE takes two values $j = l + 1/2$ ($\Delta E > 0$) and $l - 1/2$ ($\Delta E < 0$)

- For an alkali atom, the spin-orbit interaction is the ONLY weak interaction acting on the optically active electron. There are NO other (relevant) weaker interactions.

An example: alkali atoms



Spin-orbit interaction, acting on the outer 3*s* (Na) electron

For $l = 0$, then $\Delta E = 0$

For $l \neq 0$, ΔE takes two values. Each energy level is split into two components, one slightly higher (l and s parallel) and one slightly lower (l and s antiparallel)

The quantum numbers describing the state of the electron go from (n, l, s, m_l, m_s) to (n, l, s, j, m_j)

Splitting due to spin-orbit interaction

Spin-Orbit Splittings in a Number of Alkali Atoms

Element	^{3}Li	^{11}Na	^{19}K	^{37}Rb	^{55}Cs
Subshell	$2p$	$3p$	$4p$	$5p$	$6p$
Spin-orbit splitting	0.42×10^{-4} ev	21×10^{-4}	72×10^{-4}	295×10^{-4}	687×10^{-4}

Splitting of the energy levels of an electron excited to the first *p*-subshell of various alkali atoms.

It increases with Z because $\left\langle \frac{dV(r)}{dr} \right\rangle$ increases.

Quantum defect and screening

Experimental fact: the energy levels of the valence electron can be fitted by a hydrogen-like formula:

$$E_{n,l} = -R_{Na}\frac{1}{n_{eff}^2} = -R_{Na}\frac{1}{[n-\Delta(n,l)]^2}$$

$\Delta(n,l)$ is called the *quantum defect*

Table 11.2. Quantum defects $\Delta(n,l)$ for the spectra of the Na atom [from F. Richtmyer, E. Kennard, J. Cooper: *Introduction to Modern Physics*, 6th ed. (McGraw-Hill, New York 1969)]. These are empirical values

	Term	$n=3$	4	5	6	7	8
$l=0$	s	1.373	1.357	1.352	1.349	1.348	1.351
1	p	0.883	0.867	0.862	0.859	0.858	0.857
2	d	0.010	0.011	0.013	0.011	0.009	0.013
3	f	–	0.000	−0.001	−0.008	−0.012	−0.015

The quantum defects are largest for *s*–electrons, decrease with increasing l, and are quite independent of n.

They reflect the different degrees of screening of *s*, *p*, *d*, etc, electrons.

Atoms with several optically active electrons

- More typical case: Atom with a **core** of completely filled subshells surrounding the nucleus plus **several** electrons in a **partially filled subshell**. Any of these electrons can participate in the excitations leading to the optical spectrum

- The excited states are treated first by using Hartree's model which accounts for the strong interactions felt by these electrons. Then we include other **weaker** interactions.

- Remember that according to Hartree's model, the energy of each optical electron depends on the quantum numbers n, l . Then, the total energy would be that of the core (constant) plus the energies of the optically active electrons. Now, as each subshell is characterized by (n, l), then it has a degeneracy $2(2l + 1)$. (Remember the different values for m and m_s)

- There are different states with the same energy. However, this degeneracy is removed for the optical electrons.

- How?.. .....By taking into account other (weaker) interactions.

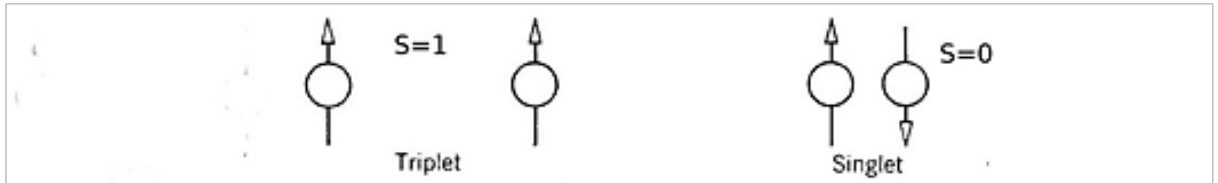
Atoms with several optically active electrons

Weaker interactions: Corrections for additional effects ignored in Hartree approx.

- Two important corrections: Residual Coulomb interaction and spin-orbit interaction
- **Residual Coulomb interaction.** It appears when dealing with atoms with two or more optically active electrons.
- The Coulomb field acting on an optically active electron includes the contribution of the other optical electrons. This is NOT spherically symmetric although Hartree potential is taken as spherically symmetric. *We have to account for these deviations.*
- To evaluate the average value of those deviations, we have to take into account the antisymmetry of the wavefunction.
- We have several (two or more) optical electrons. Given that they must be described by an anti-symmetric wavefunction, THIS FACT imposes some connexion between the relative orientation of their spins and their spatial coordinates.

Atoms with several optically active electrons

We know that the relative distance between two electrons is greater in triplet states where the spins are parallel (and so, the repulsive Coulomb energy is smaller).



A schematic illustration of the tendency for electrons in a triplet spin state to be relatively far apart, and the tendency for electrons in a singlet spin state to be relatively close together.

The residual Coulomb interaction produces a tendency for the spin angular momenta of the optically active electrons to couple in such a way that the magnitude of the total spin angular momentum S is constant and the energy is lowest for the state in which S is largest.

This coupling gives rise to a total spin

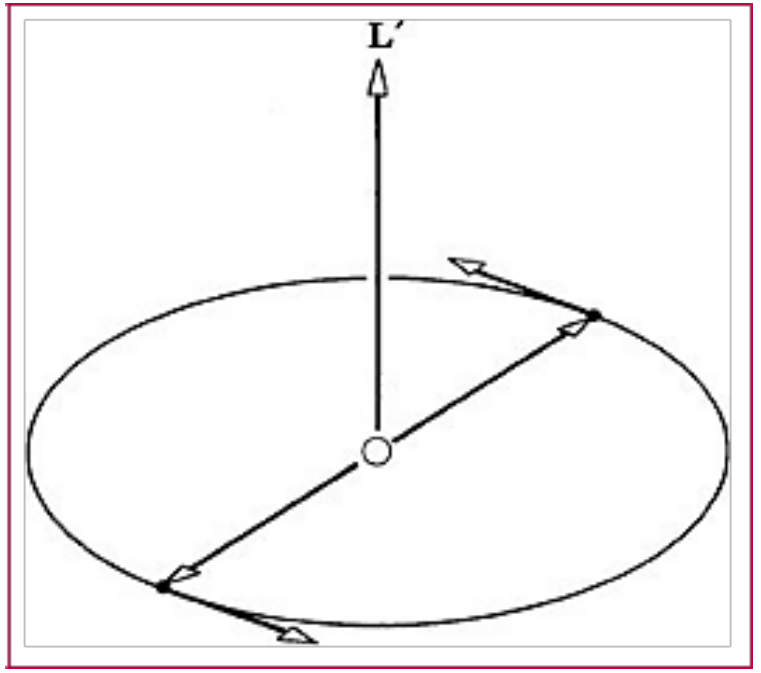
$$S = S_1 + S_2 + S_3 + \cdots + S_i + \cdots$$

Atoms with several optically active electrons

Residual Coulomb interaction

There is a tendency for the orbital angular momenta of the optical electrons to couple in such a way that their L_i are parallel.

It happens because the charge distributions of the optical electrons are not spherically symmetrical and so they exert torques on each other.



Two optically active electrons moving in the same Bohr orbit tend to remain at opposite ends of a diameter so as to minimize their Coulomb repulsion. As a result, their orbital angular momenta tend to couple in such a way as to yield a maximum total orbital angular momentum.

Atoms with several optically active electrons

Another effect of the residual Coulomb interaction is that it induces an effective force on the optical electrons such that it tends to align the orbital angular momenta of the electrons in the outer shell, i.e. it induces a coupling among the orbital angular momenta of the electrons in the outer shell so as to give a total orbital angular momentum

$$L = L_1 + L_2 + L_3 + \cdots + L_i + \cdots$$

which remains constant and the energy is lowest for the state with L largest

Atoms with several optically active electrons

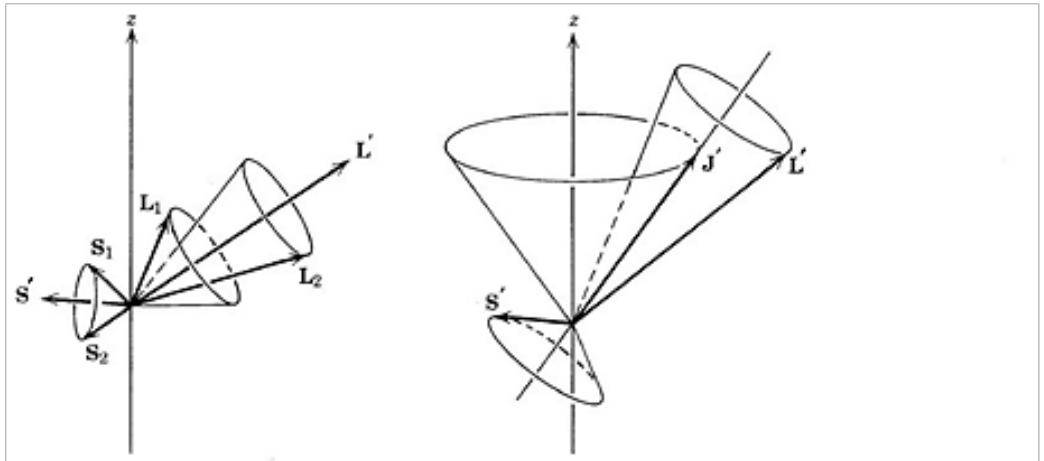
Spin-orbit interaction

- There is an additional coupling between the total orbital angular momenta and total spin angular momenta.
- The magnetic dipole moments associated to the orbital and spin angular momenta induce an additional coupling between the total L and total S, giving rise to a total angular momentum

$$J = L + S$$

- Except for atoms with large Z, the residual Coulomb interaction is more important and therefore it is treated first. Moreover, the coupling of the spins comes first and then the coupling of the orbital angular momenta
- Finally, the weaker spin-orbit interaction is included. This interaction couples the total *S* and *L*, in such a way that $J = L + S$ is constant. But *J* can take different values with associated different energies. The state with minimum J has the lowest energy.
- This procedure is called **LS coupling**

Atoms with several optically active electrons

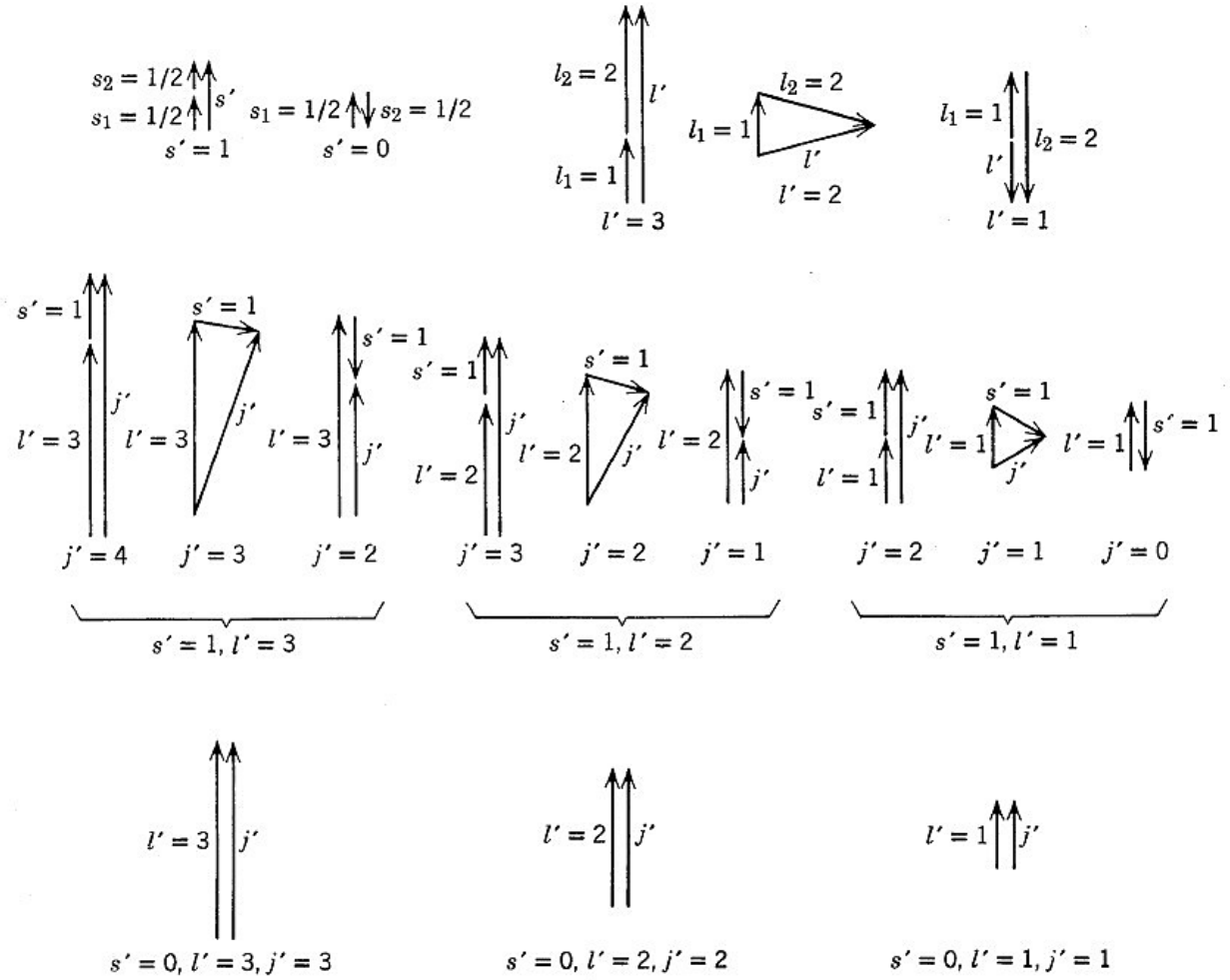


The coupling of various angular momentum vectors in a typical *LS* coupling state of minimum energy. *Left:* The orbital angular momenta $\mathbf{L}_1$ and $\mathbf{L}_2$ of the two electrons precess rapidly about their vector sum $\mathbf{L}'$. Similarly, their spins $\mathbf{S}_1$ and $\mathbf{S}_2$ precess rapidly about their sum $\mathbf{S}'$. *Right:* The total orbital angular momentum $\mathbf{L}'$ and the total spin angular momentum $\mathbf{S}'$ precess slowly about their sum $\mathbf{J}'$, the total angular momentum. Finally, $\mathbf{J}'$ can be found anywhere on a cone symmetrical about the z axis.

The orbital angular momenta $\mathbf{L}_1$ and $\mathbf{L}_2$ precess rapidly about their sum $\mathbf{L}'$ because their coupling is also relatively strong, and $\mathbf{L}'$ also has its maximum possible magnitude (corresponding to $l' = 3$).

electrons with quantum numbers $l_1 = 1$, $s_1 = 1/2$, and $l_2 = 2$, $s_2 = 1/2$. The spin angular momenta $\mathbf{S}_1$ and $\mathbf{S}_2$ precess about their sum $\mathbf{S}'$, and $\mathbf{S}'$ has its maximum possible magnitude (corresponding to $s' = 1$).

Atoms with several optically active electrons



Vector addition diagrams for the quantum numbers $l_1 = 1$, $s_1 = 1/2$; $l_2 = 2$, $s_2 = 1/2$.

Atoms with several optically active electrons

Within this LS coupling scheme, the description of the states associated to the system of the outer electrons is made in terms: L^2, S^2, J^2, J_z which are a set of compatible observables.

Now, the states are described by the notation $|l, s, j, m_j\rangle$.

The associated levels are denoted as $^{2s+1}l_j$

$l = 0\ (S), 1\ (P), 2\ (D), 3\ (F), \ldots$

Now, the state of the atom is characterized (the ionic core is considered inert) by the quantum numbers l, s, j, m_j

States with the same values of (l, s) are named **terms**

States of a term (same l, s) but different j are called **multiplet**

The total number of different *j*-values associated to a term (*l*, s) is called the *multiplicity* of the term

Each term has, in principle, a different energy.

Atoms with several optically active electrons

Let us consider several examples of configurations. How to find all the terms?

Two optical electrons, one in the outer *ns* subshell and the other in the excited state *n's*

(Example. Be: $1s^2$, $2s^1$, $3s^1$)

We have $l_1 = 0,\ s_1 = 1/2$ and $l_2 = 0,\ s_2 = 1/2$

Then $l = 0$ and $s = 0, 1$

So, we have the terms ${}^3S_1,\ {}^1S_0$

With the old notation we had 2 (*ns*) states X 2 (*n's*) states = 4 states (degenerate)

Notice that if both electrons were in the same *ns* subshell (i.e. Be: $1s^2$, $2s^2$), then there would be fewer terms, namely (n,0,0, +1/2) and (n,0,0, -1/2). This is due to the exclusion principle. Then

$m_s = 0,\ s = 0$ (and $l = 0$) $\rightarrow$ ${}^1S_0$

Atoms with several optically active electrons

Two optical electrons, one in the outer *np* subshell and the other in the excited state *n′p*

(Example. C: $1s^2$, $2s^2$, $2p^1$, $3p^1$)

We have $l_1 = 1, s_1 = 1/2$ and $l_2 = 1, s_2 = 1/2$

Then $l = 2, 1, 0$ and $s = 0, 1$

We have the terms

$(2,1) \rightarrow {}^3D_{3,2,1}$	(15 states)	$(2,0) \rightarrow {}^1D_2$	(5 states)
$(1,1) \rightarrow {}^3P_{2,1,0}$	(9 states)	$(1,0) \rightarrow {}^1P_1$	(3 states)
$(0,1) \rightarrow {}^3S_1$	(3 states)	$(0,0) \rightarrow {}^1S_0$	(1 states)

Taking into account that for each j there are 2j+1 states, we have a total of 36 states.

With the old notation we had 6 (*np*) states X 6 (*n′p*) states = 36 states

Atoms with several optically active electrons

Two optical electrons, in configuration $3d^1\ 4p^1$

We have $l_1 = 2, s_1 = 1/2$ and $l_2 = 1, s_2 = 1/2$

Then $l = 3, 2, 1$ and $s = 0, 1$

Finally, we have ${}^1P_1$, ${}^3P_{2,1,0}$, ${}^1D_2$, ${}^3D_{3,2,1}$, ${}^1F_3$ and ${}^3F_{4,3,2}$.

It makes a total of 60 states.

In the old notation we had, 10 ($3d^1$) x 6 ($4p^1$) = 60 states

Atoms with several optically active electrons

When the outer electrons are in the same subshell, we have to take into account the exclusion principle.

This means that there are fewer available states

For example : Two optical electrons, in configuration n*p*, n*p*

How many states are available? $C_6^2 = 15$ configurations

These are $^1S_0$, $^1D_2$, $^3P_{2,1,0}$

(see Sanchez del Rio, pag 611).

Notice that the number of allowed states is much smaller than in the case of optical electrons in configuration n'*p*, n*p*

Atoms with several optically active electrons

Two optical electrons, in configuration n*p*, n*p*

Diferentes estados de un acoplo Russell-Saunders

	m_{ℓ_1}	1		0		−1	
m_{ℓ_2}	m_{s_1} m_{s_2}	1/2	−1/2	1/2	−1/2	1/2	−1/2
1	1/2	m_s m_ℓ m_j	0 2 2	1 1 2	0 1 1	1 0 1	0 0 0
	−1/2			0 1 1	−1 1 0	0 0 0	−1 0 −1
0	1/2				0 0 0	1 −1 0	0 −1 −1
	−1/2					0 −1 −1	−1 −1 −2
−1	1/2						0 −2 −2
	−1/2						

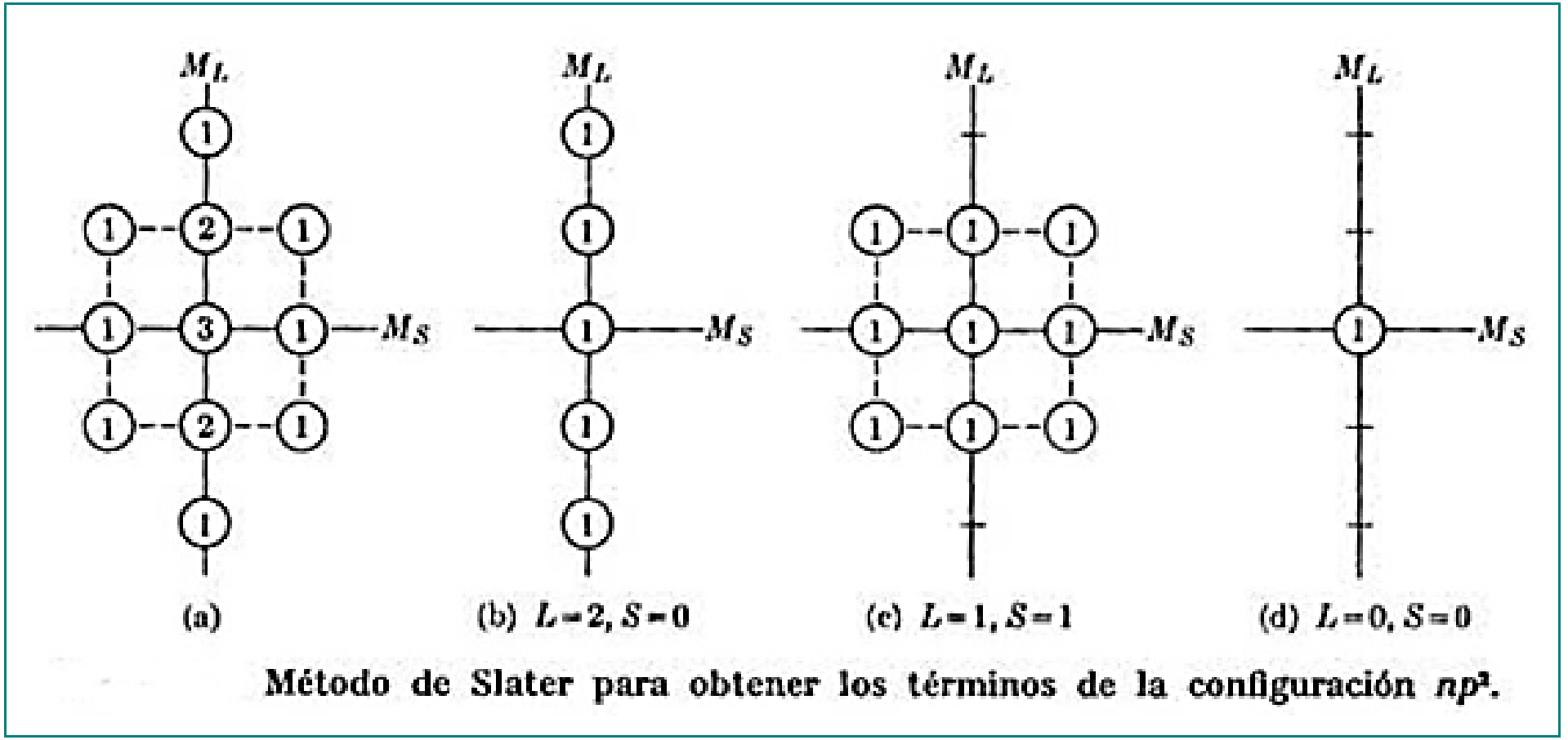
Atoms with several optically active electrons

$M_L = m_{l1} + m_{l2}$ $M_S = m_{s1} + m_{s2}$

Posibles valores M_L y M_S para la configuración np^2

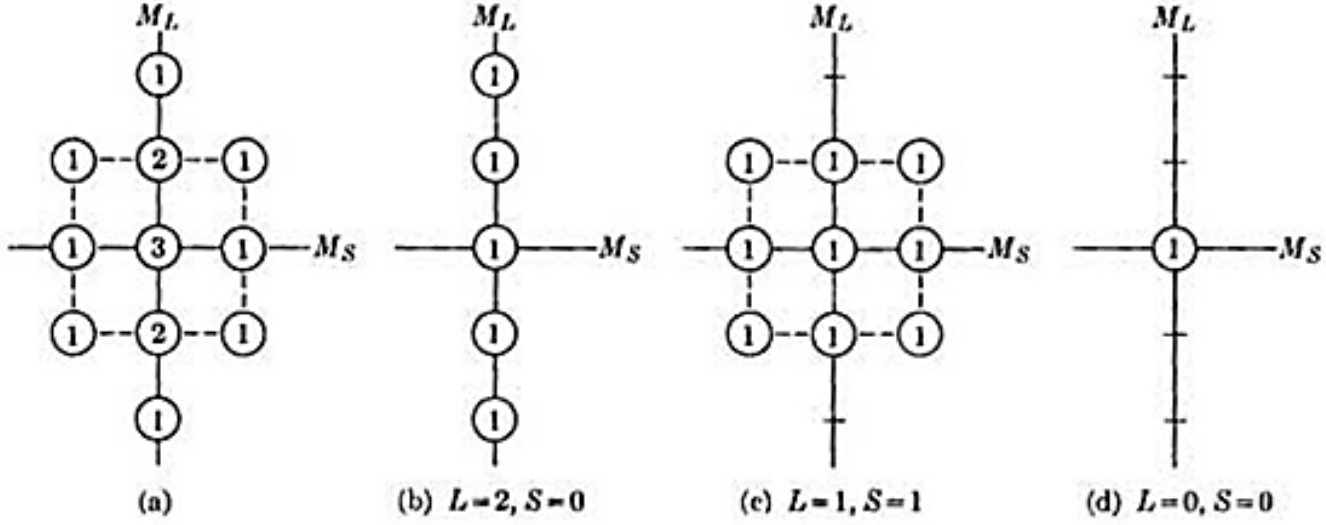
M_L \ M_S	− 1	0	+ 1
+ 2		(1, +) (1, —)	
+ 1	(1, —) (0, —)	(1, +) (0, —) (1, —) (0, +)	(1, +) (0, +)
0	(1, —) (— 1, —)	(1, +) (— 1, —) (0, +) (0, —) (1, —) (— 1, +)	(1, +) (— 1, +)
— 1	(0, —) (— 1, —)	(0, +) (— 1, —) (0, —) (— 1, +)	(0, +) (— 1, +)
— 2		(— 1, +) (— 1, —)	

Atoms with several optically active electrons



Método de Slater para obtener los términos de la configuración np^2.

Atoms with several optically active electrons



Método de Slater para obtener los términos de la configuración np^2.

Posibles valores M_L y M_S para la configuración np^2

M_L \ M_S	-1	0	$+1$
$+2$		$(1,+)\,(1,-)$	
$+1$	$(1,-)\,(0,-)$	$(1,+)\,(0,-)$ $(1,-)\,(0,+)$	$(1,+)\,(0,+)$
0	$(1,-)\,(-1,-)$	$(1,+)\,(-1,-)$ $(0,+)\,(0,-)$ $(1,-)\,(-1,+)$	$(1,+)\,(-1,+)$
-1	$(0,-)\,(-1,-)$	$(0,+)\,(-1,-)$ $(0,-)\,(-1,+)$	$(0,+)\,(-1,+)$
-2		$(-1,+)\,(-1,-)$	

Atoms with several optically active electrons

Terms corresponding to equivalent electrons

ns^0	^1S																			
ns^1				^2S																
ns^2	^1S																			
np^0	^1S																			
np^1				^2P																
np^2	^1S,	^1D					^3P													
np^3				^2P,	^2D					^4S										
np^4	^1S,	^1D					^3P													
np^5				^2P																
np^6	^1S																			
nd^0	^1S,																			
nd^1				^2D																
nd^2	^1S,	^1D,	^1G							^3P,	^3F									
nd^3				^2D,	^2P,	^2D,	^2F,	^2G,	^2H					^4P,	^3F					
nd^4	^1S,	^1D,	^1G,	^1S,	^1D,	^1G,	^1H,	^1I		^3P,	^3F,	^3P,	^3D,	^3F,	^3G,	^3H		^5D		
nd^5				^2D,	^2P,	^2D,	^2F,	^2G,	^2H,	^2S,	^2D,	^2F,	^2G,	^2I		^4P,	^2F,	^4D,	^4G		^6S
nd^6	^1S,	^1D,	^1G,	^1S,	^1D,	^1G,	^1H,	^1I		^3P,	^3F,	^3P,	^3D,	^3F,	^3G,	^3H		^5D		
nd^7				^2D,	^2P,	^2D,	^2F,	^2G,	^2H					^4P,	^4F					
nd^8	^1S,	^1D,	^1G							^3P,	^3F									
nd^9				^2D																
nd^{10}	^1S																			

Atoms with several optically active electrons

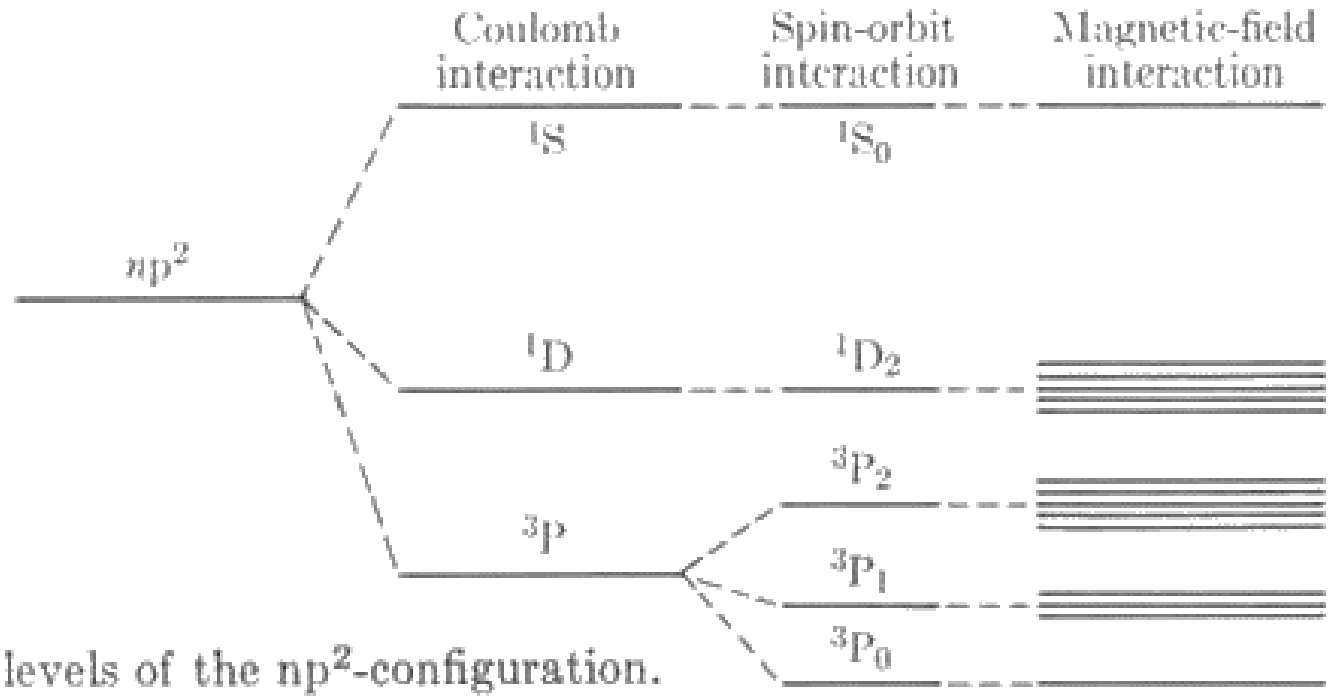
Once the allowed spectral terms have been found, now, we have to position them according to their energies.

There are some empirical rules (Hund)

- Among all the possible terms, those with maximum s value, have the lowest energy.
- Among all the terms with the same s value, those with the maximum l, have the lowest energy
- Among all the terms with the same s and l values, those with the minimum j, have the lowest energy

From here, we can now analyze all the possible optical transitions.

Atoms with several optically active electrons

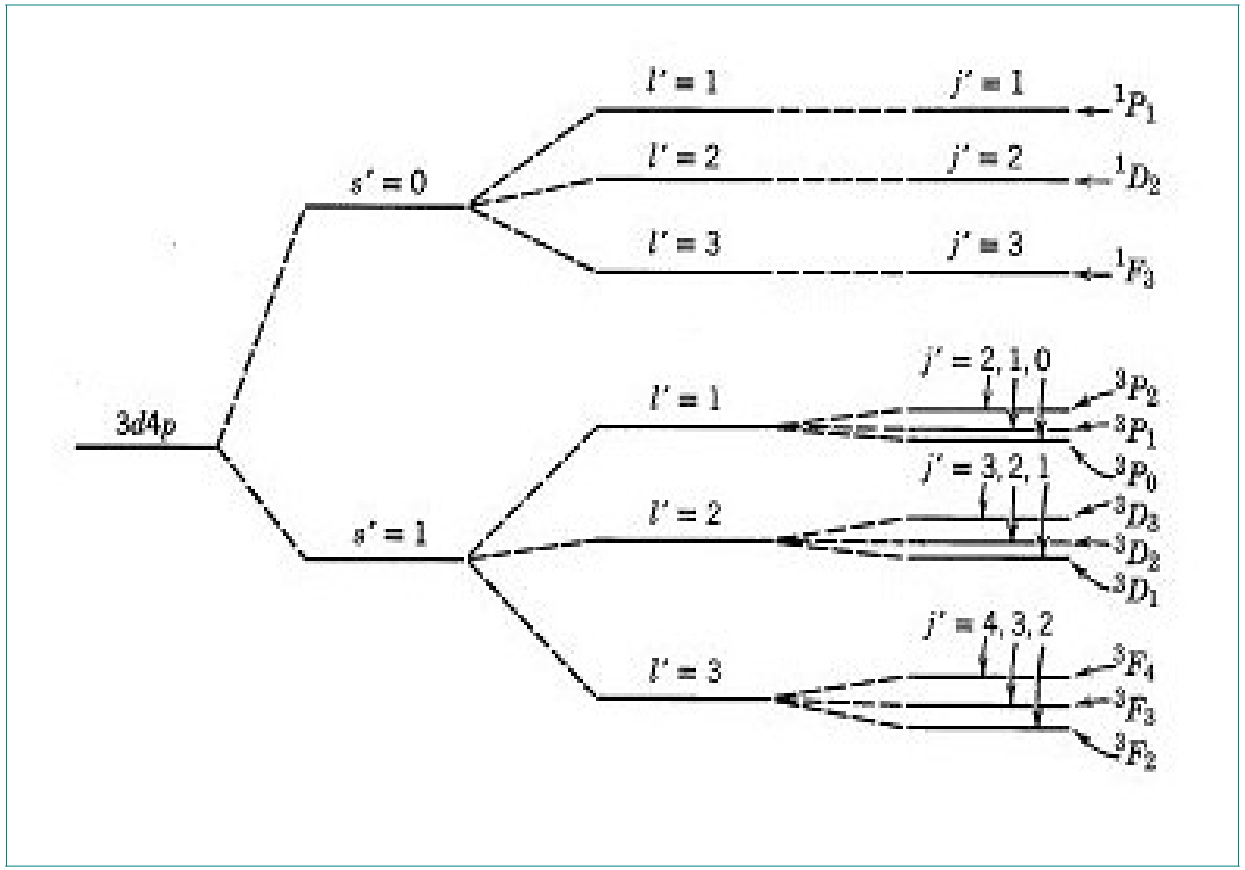


Energy levels of the np^2-configuration.

Atoms with several optically active electrons

Splitting of the energy levels in a typical LS coupling configuration

$3d^1 - 4p^1$ configuration $^1P_1$, $^3P_{2,1,0}$, $^1D_2$, $^3D_{3,2,1}$, $^1F_3$, $^3F_{4,3,2}$.



$$\overline{\Delta E} = K[j'(j'+1) - l'(l'+1) - s'(s'+1)]$$

This equation predicts the expectation value of the interaction energy of the total spin and orbital angular momentum vectors $\mathbf{S}'$ and $\mathbf{L}'$, providing *LS* coupling is valid so that these vectors are meaningful. The quantity K is not simply proportional to a term like $\overline{(1/r)\,dV(r)/dr}$, as might be expected from earlier applications of (8-35), because the potential is more complicated in the present situation. However, K does have the same value for all the energy levels of a so-called *multiplet*;

Atoms with several optically active electrons

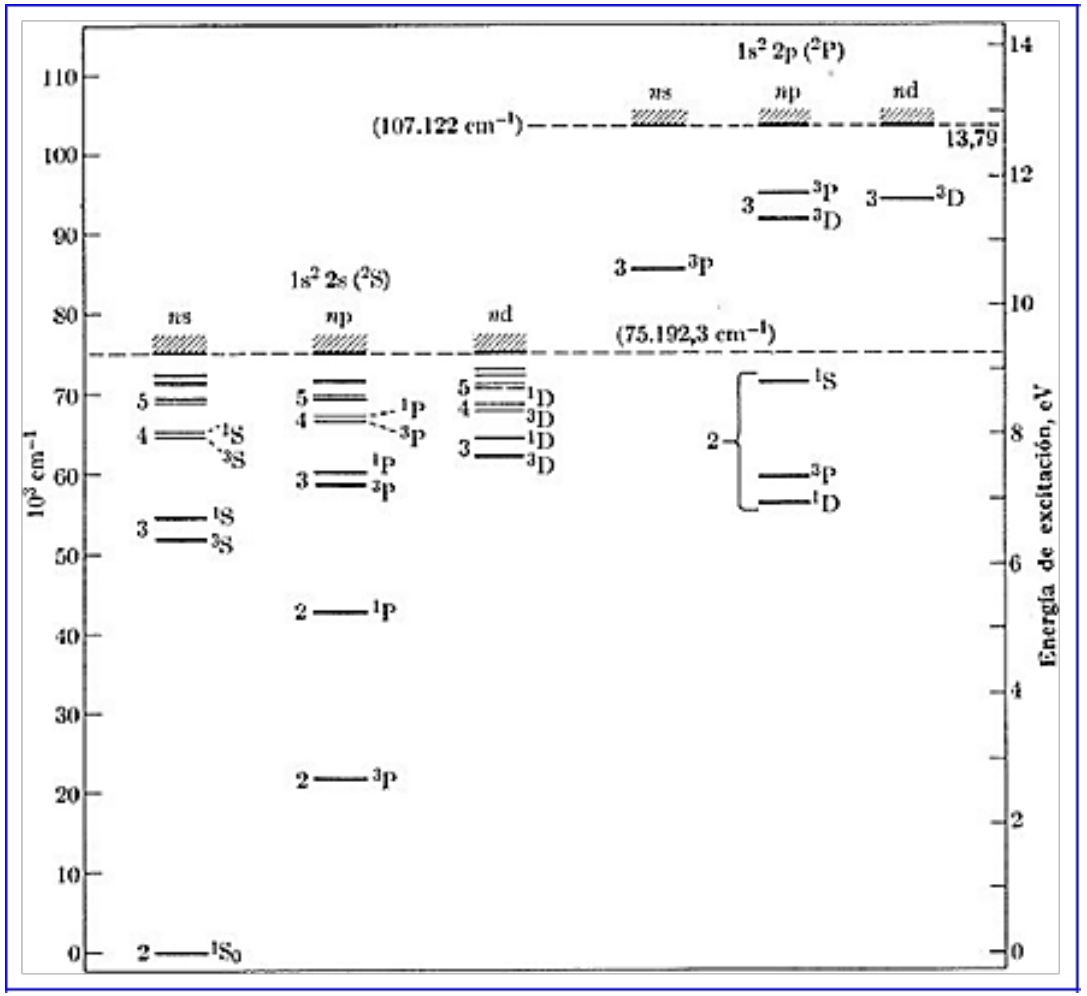
- When a subshell is completely filled, the only allowed state is that one in which the total spin angular momentum, the total orbital angular momentum and the total angular momentum are all zero.

- Consequently, there is zero total angular momentum in a completely filled subshell. → It has NO net magnetic dipole moment.

- Only those few electrons in unfilled subshells are involved in its interaction with external magnetic fields.

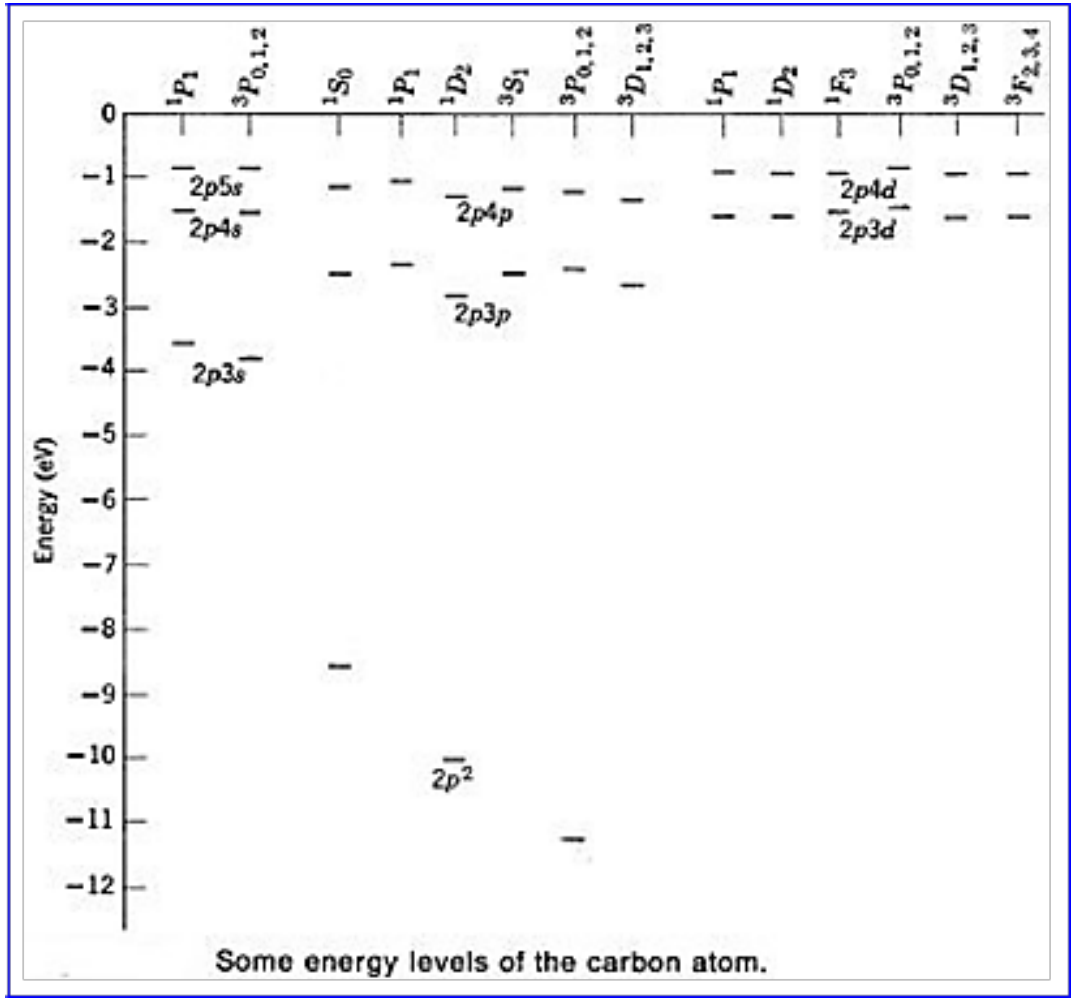
Atoms with several optically active electrons

Energy levels of beryllium atom. Ground state is $1s^2\ 2s^2$



Atoms with several optically active electrons

Energy levels of carbon atom. Ground state is $1s^2\ 2s^2\ 2p^2$



Some energy levels of the carbon atom.

$2p^2$ is the lowest energy configuration

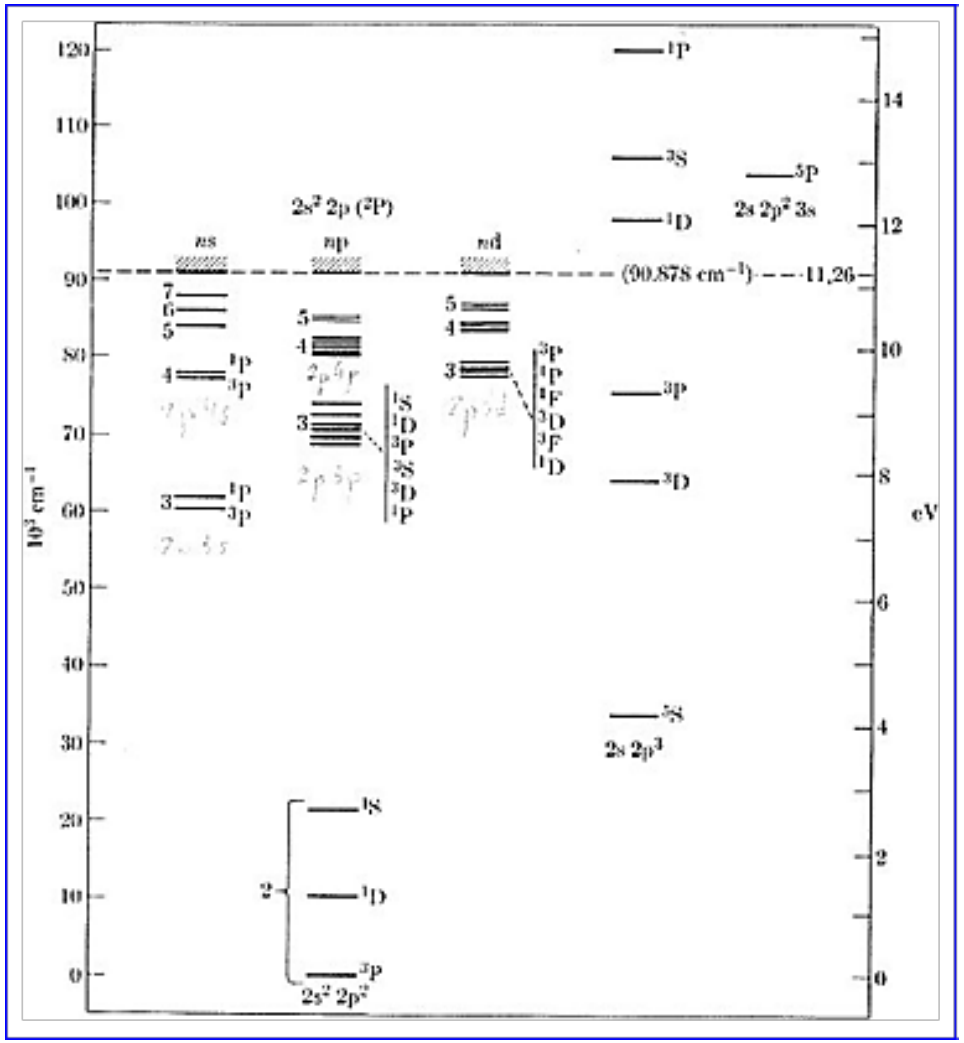
In the other configurations one electron remains in $2p$ and the other goes to a subshell of higher energy.

In the $2p^2$ configuration the lowest energy is $^3P_0$.

Sometimes it is written that the ground state of the atom is $2p^2\ ^3P_0$.

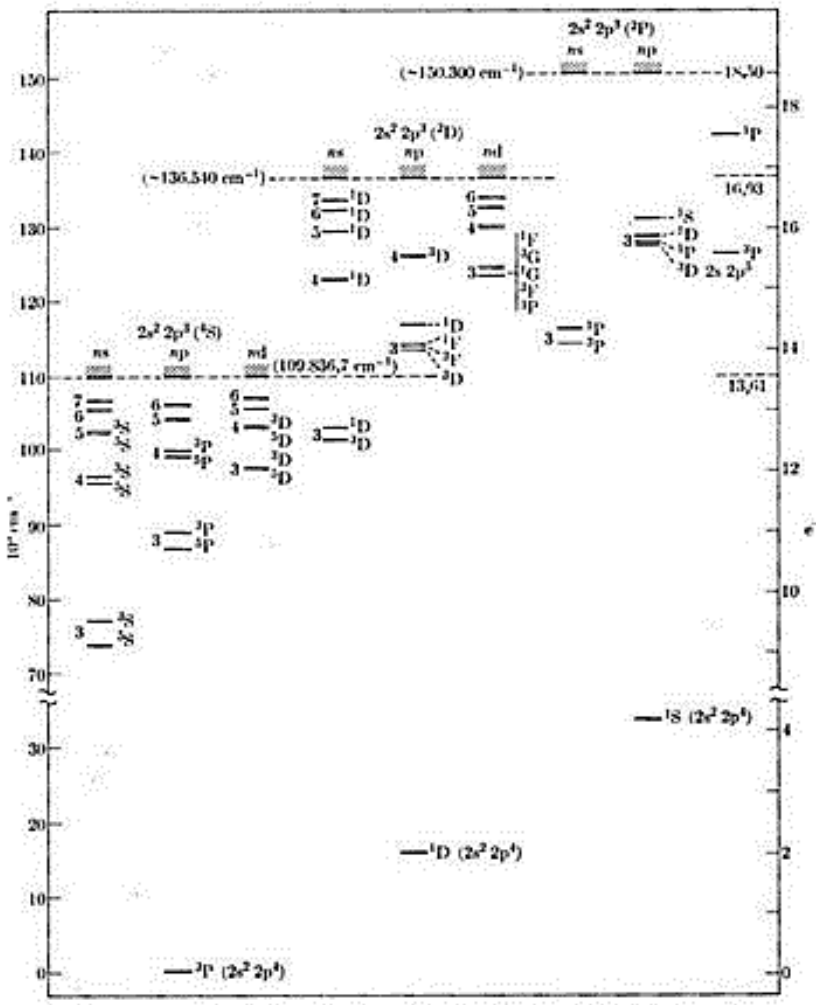
Atoms with several optically active electrons

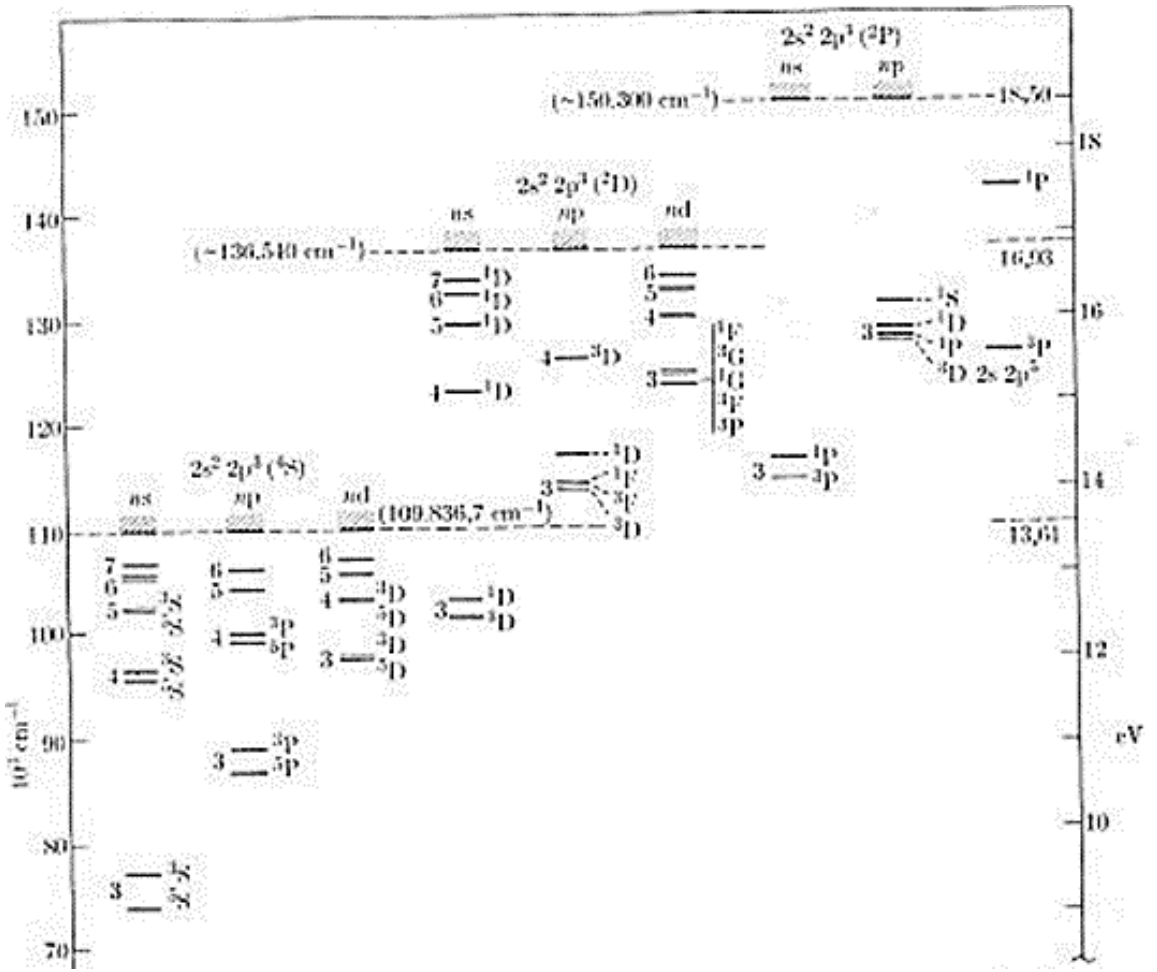
Energy levels of carbon atom. Ground state is $1s^2\ 2s^2\ 2p^2$



Atoms with several optically active electrons

Energy levels of oxygen atom. Ground state is $1s^2\ 2s^2\ 2p^4$





Atoms with several optically active electrons

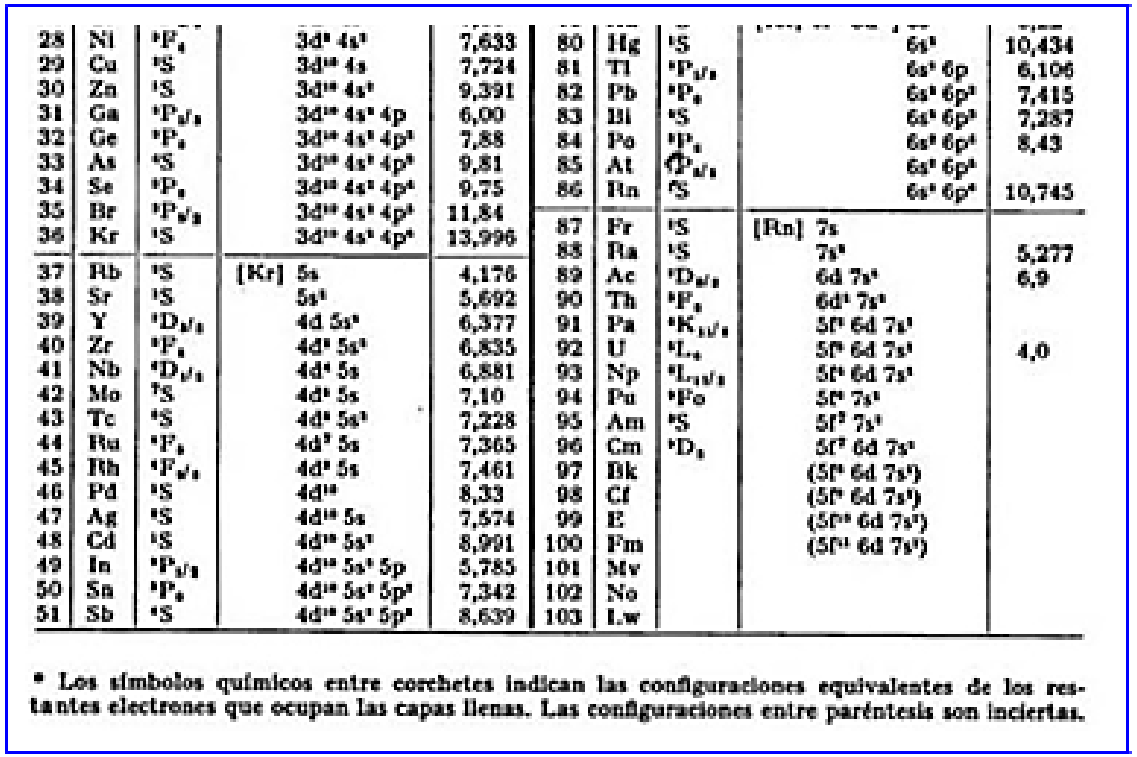
Electronic configuration of atoms

Z	Element	Symbol	Configuration	Term
1	Hydrogen	H	$1s^1$	$^2S_{1/2}$
2	Helium	He	$1s^2$	$^1S_0$
3	Lithium	Li	$1s^2 2s^1$	$^2S_{1/2}$
4	Beryllium	Be	$1s^2 2s^2$	$^1S_0$
5	Boron	B	$1s^2 2s^2 2p^1$	$^2P_{1/2}$
6	Carbon	C	$1s^2 2s^2 2p^2$	$^3P_0$
7	Nitrogen	N	$1s^2 2s^2 2p^3$	$^4S_{3/2}$
8	Oxygen	O	$1s^2 2s^2 2p^4$	$^3P_2$
9	Fluorine	F	$1s^2 2s^2 2p^5$	$^2P_{3/2}$
10	Neon	Ne	$1s^2 2s^2 2p^6$	$^1S_0$
11	Sodium	Na	$[Ne]3s^1$	$^2S_{1/2}$
12	Magnesium	Mg	$[Ne]3s^2$	$^1S_0$
13	Aluminum	Al	$[Ne]3s^2 3p^1$	$^2P_{1/2}$
14	Silicon	Si	$[Ne]3s^2 3p^2$	$^3P_0$
15	Phosphorus	P	$[Ne]3s^2 3p^3$	$^4S_{3/2}$
16	Sulfur	S	$[Ne]3s^2 3p^4$	$^3P_2$
17	Chlorine	Cl	$[Ne]3s^2 3p^5$	$^2P_{3/2}$
18	Argon	Ar	$[Ne]3s^2 3p^6$	$^1S_0$
19	Potassium	K	$[Ar]4s^1$	$^2S_{1/2}$
20	Calcium	Ca	$[Ar]4s^2$	$^1S_0$
21	Scandium	Sc	$[Ar]3d^1 4s^2$	$^2D_{3/2}$
22	Titanium	Ti	$[Ar]3d^2 4s^2$	$^3F_2$
23	Vanadium	V	$[Ar]3d^3 4s^2$	$^4F_{3/2}$
24	Chromium	Cr	$[Ar]3d^5 4s^1$	$^7S_3$
25	Manganese	Mn	$[Ar]3d^5 4s^2$	$^6S_{5/2}$
26	Iron	Fe	$[Ar]3d^6 4s^2$	$^5D_4$
27	Cobalt	Co	$[Ar]3d^7 4s^2$	$^4F_{9/2}$
28	Nickel	Ni	$[Ar]3d^8 4s^2$	$^3F_4$
29	Copper	Cu	$[Ar]3d^{10} 4s^1$	$^2S_{1/2}$
30	Zinc	Zn	$[Ar]3d^{10} 4s^2$	$^1S_0$
31	Gallium	Ga	$[Ar]3d^{10} 4s^2 4p^1$	$^2P_{1/2}$
32	Germanium	Ge	$[Ar]3d^{10} 4s^2 4p^2$	$^3P_0$
33	Arsenic	As	$[Ar]3d^{10} 4s^2 4p^3$	$^4S_{3/2}$
34	Selenium	Se	$[Ar]3d^{10} 4s^2 4p^4$	$^3P_2$
35	Bromine	Br	$[Ar]3d^{10} 4s^2 4p^5$	$^2P_{3/2}$
36	Krypton	Kr	$[Ar]3d^{10} 4s^2 4p^6$	$^1S_0$

An example: Other multielectron atoms

TABLA 4-1 Configuración electrónica de los átomos *

Z	Símbolo	Estado fundamental	Configuración fundamental	Energía de ionización, eV	Z	Símbolo	Estado fundamental	Configuración fundamental	Energía de ionización, eV
1	H	$^2S_{1/2}$	1s	13,595	52	Te	$^3P_2$	$4d^{10}\,5s^2\,5p^4$	9,01
2	He	$^1S_0$	$1s^2$	24,581	53	I	$^2P_{3/2}$	$4d^{10}\,5s^2\,5p^5$	10,454
					54	Xe	$^1S_0$	$4d^{10}\,5s^2\,5p^6$	12,127
3	Li	$^2S_{1/2}$	[He] 2s	5,390					
4	Be	$^1S_0$	$2s^2$	9,320	55	Cs	$^2S_{1/2}$	[Xe] 6s	3,893
5	B	$^2P_{1/2}$	$2s^2\,2p$	8,296	56	Ba	$^1S_0$	$6s^2$	5,210
6	C	$^3P_0$	$2s^2\,2p^2$	11,256	57	La	$^2D_{3/2}$	5d 6s²	5,61
7	N	$^4S_{3/2}$	$2s^2\,2p^3$	14,545	58	Ce	$^1G_4$	4f 5d 6s²	6,54
8	O	$^3P_2$	$2s^2\,2p^4$	13,614	59	Pr	$^4I_{9/2}$	$4f^3\,6s^2$	5,48
9	F	$^2P_{3/2}$	$2s^2\,2p^5$	17,418	60	Nd	$^5I_4$	$4f^4\,6s^2$	5,51
10	Ne	$^1S_0$	$2s^2\,2p^6$	21,559	61	Pm	$^6H_{5/2}$	$4f^5\,6s^2$	
					62	Sm	$^7F_0$	$4f^6\,6s^2$	5,6
11	Na	$^2S_{1/2}$	[Ne] 3s	5,138	63	Eu	$^8S_{7/2}$	$4f^7\,6s^2$	5,67
12	Mg	$^1S_0$	$3s^2$	7,644	64	Gd	$^9D_2$	$4f^7\,5d\,6s^2$	6,16
13	Al	$^2P_{1/2}$	$3s^2\,3p$	5,984	65	Tb	$^6H_{15/2}$	$4f^9\,6s^2$	6,74
14	Si	$^3P_0$	$3s^2\,3p^2$	8,149	66	Dy	$^5I_8$	$4f^{10}\,6s^2$	6,82
15	P	$^4S_{3/2}$	$3s^2\,3p^3$	10,484	67	Ho	$^4I_{15/2}$	$4f^{11}\,6s^2$	
16	S	$^3P_2$	$3s^2\,3p^4$	10,357	68	Er	$^3H_6$	$4f^{12}\,6s^2$	
17	Cl	$^2P_{3/2}$	$3s^2\,3p^5$	13,01	69	Tm	$^2F_{7/2}$	$4f^{13}\,6s^2$	
18	Ar	$^1S_0$	$3s^2\,3p^6$	15,755	70	Yb	$^1S_0$	$4f^{14}\,6s^2$	6,22
19	K	$^2S_{1/2}$	[Ar] 4s	4,339	71	Lu	$^2D_{3/2}$	$4f^{14}\,5d\,6s^2$	6,15
20	Ca	$^1S_0$	$4s^2$	6,111	72	Hf	$^3F_2$	$4f^{14}\,5d^2\,6s^2$	7,0
21	Sc	$^2D_{3/2}$	$3d\,4s^2$	6,54	73	Ta	$^4F_{3/2}$	$4f^{14}\,5d^3\,6s^2$	7,88
22	Ti	$^3F_2$	$3d^2\,4s^2$	6,83	74	W	$^5D_0$	$4f^{14}\,5d^4\,6s^2$	7,98
23	V	$^4F_{3/2}$	$3d^3\,4s^2$	6,74	75	Re	$^6S_{5/2}$	$4f^{14}\,5d^5\,6s^2$	7,87
24	Cr	$^7S_3$	$3d^5\,4s$	6,764	76	Os	$^5D_4$	$4f^{14}\,5d^6\,6s^2$	8,7
25	Mn	$^6S_{5/2}$	$3d^5\,4s^2$	7,432	77	Ir	$^4F_{9/2}$	$4f^{14}\,5d^7\,6s^2$	9,2
26	Fe	$^5D_4$	$3d^6\,4s^2$	7,87	78	Pt	$^3D_3$	$4f^{14}\,5d^9\,6s$	8,88
27	Co	$^4F_{9/2}$	$3d^7\,4s^2$	7,86	79	Au	$^2S_{1/2}$	[Xe, $4f^{14}\,5d^{10}$] 6s	9,22
28	Ni	$^3F_4$	$3d^8\,4s^2$	7,633	80	Hg	$^1S_0$	$6s^2$	10,434
29	Cu	$^2S_{1/2}$	$3d^{10}\,4s$	7,724	81	Tl	$^2P_{1/2}$	$6s^2\,6p$	6,106
30	Zn	$^1S_0$	$3d^{10}\,4s^2$	9,391	82	Pb	$^3P_0$	$6s^2\,6p^2$	7,415



28	Ni	$^3F_4$		$3d^8\,4s^2$	7,633	80	Hg	$^1S_0$	$6s^2$ 10,434
29	Cu	$^2S_{1/2}$		$3d^{10}\,4s$	7,724	81	Tl	$^2P_{1/2}$	$6s^2\,6p$ 6,106
30	Zn	$^1S_0$		$3d^{10}\,4s^2$	9,391	82	Pb	$^3P_0$	$6s^2\,6p^2$ 7,415
31	Ga	$^2P_{1/2}$		$3d^{10}\,4s^2\,4p$	6,00	83	Bi	$^4S_{3/2}$	$6s^2\,6p^3$ 7,287
32	Ge	$^3P_0$		$3d^{10}\,4s^2\,4p^2$	7,88	84	Po	$^3P_2$	$6s^2\,6p^4$ 8,43
33	As	$^4S_{3/2}$		$3d^{10}\,4s^2\,4p^3$	9,81	85	At	$^2P_{3/2}$	$6s^2\,6p^5$
34	Se	$^3P_2$		$3d^{10}\,4s^2\,4p^4$	9,75	86	Rn	$^1S_0$	$6s^2\,6p^6$ 10,745
35	Br	$^2P_{3/2}$		$3d^{10}\,4s^2\,4p^5$	11,84				
36	Kr	$^1S_0$		$3d^{10}\,4s^2\,4p^6$	13,996	87	Fr	$^2S_{1/2}$	[Rn] 7s
						88	Ra	$^1S_0$	$7s^2$ 5,277
37	Rb	$^2S_{1/2}$	[Kr] 5s		4,176	89	Ac	$^2D_{3/2}$	6d 7s² 6,9
38	Sr	$^1S_0$	$5s^2$		5,692	90	Th	$^3F_2$	$6d^2\,7s^2$
39	Y	$^2D_{3/2}$	4d 5s²		6,377	91	Pa	$^4K_{11/2}$	$5f^2\,6d\,7s^2$
40	Zr	$^3F_2$	$4d^2\,5s^2$		6,835	92	U	$^5L_6$	$5f^3\,6d\,7s^2$ 4,0
41	Nb	$^4D_{1/2}$	$4d^4\,5s$		6,881	93	Np	$^6L_{11/2}$	$5f^4\,6d\,7s^2$
42	Mo	$^7S_3$	$4d^5\,5s$		7,10	94	Pu	^7Fo	$5f^6\,7s^2$
43	Tc	$^6S_{5/2}$	$4d^5\,5s^2$		7,228	95	Am	$^8S_{7/2}$	$5f^7\,7s^2$
44	Ru	$^5F_5$	$4d^7\,5s$		7,365	96	Cm	$^9D_2$	$5f^7\,6d\,7s^2$
45	Rh	$^4F_{9/2}$	$4d^8\,5s$		7,461	97	Bk		(5f⁸ 6d 7s²)
46	Pd	$^1S_0$	$4d^{10}$		8,33	98	Cf		(5f¹⁰ 7s²)
47	Ag	$^2S_{1/2}$	$4d^{10}\,5s$		7,574	99	E		(5f¹¹ 6d 7s²)
48	Cd	$^1S_0$	$4d^{10}\,5s^2$		8,991	100	Fm		(5f¹² 6d 7s²)
49	In	$^2P_{1/2}$	$4d^{10}\,5s^2\,5p$		5,785	101	Mv		
50	Sn	$^3P_0$	$4d^{10}\,5s^2\,5p^2$		7,342	102	No		
51	Sb	$^4S_{3/2}$	$4d^{10}\,5s^2\,5p^3$		8,639	103	Lw		

* Los símbolos químicos entre corchetes indican las configuraciones equivalentes de los restantes electrones que ocupan las capas llenas. Las configuraciones entre paréntesis son inciertas.

Notation

Optical Transitions. Selection Rules

Consider a one-electron atom. *(see Sanchez del Rio, pag 560)*

Suppose that the atom is subjected to an external electromagnetic field $\vec{E}(\vec{r},t)$ In a first order approximation, this interaction is described by a term

$$W = (-e)\vec{r}\cdot\vec{E}$$

Therefore the new electronic hamiltonian will be

$$H = H_0 + W$$

The electronic state is changed, and it can be evaluated by solving SE: $i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$

To solve it, we try solutions $|\psi(t)\rangle = \sum_n c_n(t)\,|\phi_n\rangle$

where $|\phi_n\rangle$ are the stationary states associated to the hamiltonian H_0.

We know that for $t = 0$ $|\psi(t=0)\rangle = |\phi_i\rangle$ which is the initial state before the interaction.

This means that $c_i(t=0) = 1$ and all the other coefficients are zero.

Optical Transitions. Selection Rules

We want to evaluate the probability of a transition from the initial stationary state $|\phi_i\rangle$ to another, final, stationary state $|\phi_f\rangle$

It can be shown that this probability is proportional to $|c_f(t)|^2$ with

$$|c_f(t)|^2 \propto |\langle\phi_f|W|\phi_i\rangle|^2$$

Therefore, it is important to analize these matrix elements, because when they are zero, those transitions will be forbidden.

Moreover, we know that when an atom is excited to a higher energy level (i.e. collisions), the atom will return to its ground state by making transitions to lower energy levels.

However, not all possible transitions take place. Only those corresponding to transitions between energy levels which satisfy some *selection rules*.

Optical Transitions. Selection Rules

The forbidden transitions are those for which the matrix element

$$\int d\vec{r}\,\phi_f^*(\vec{r})(-e)\vec{r}\phi_i(\vec{r})$$

is zero.

Let us write

$$\phi_f(\vec{r}) \longrightarrow |n', l', s', m_{l'}, m_{s'}\rangle$$

$$\phi_i(\vec{r}) \longrightarrow |n, l, s, m_l, m_s\rangle$$

So, we have to evaluate

$$\langle n', l', s', m_{l'}, m_{s'}|\,\vec{r}\,|n, l, s, m_l, m_s\rangle$$

which amounts to three terms

$$\langle n', l', s', m_{l'}, m_{s'}|\,\mathrm{x}\,|n, l, s, m_l, m_s\rangle \qquad \langle n', l', s', m_{l'}, m_{s'}|\,\mathrm{y}\,|n, l, s, m_l, m_s\rangle \qquad \langle n', l', s', m_{l'}, m_{s'}|\,\mathrm{z}\,|n, l, s, m_l, m_s\rangle$$

Optical Transitions. Selection Rules

Remember that $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$

Change variables $\{x, y, z\}$ to $\{\xi, \eta, z\}$, where $\xi = x + iy$, $\eta = x - iy$

First, notice that $\langle n', l', m_{l'} | \vec{r} | n, l, m_l \rangle \langle s', m_{s'} | s, m_s \rangle$

which provides a first condition, namely $\Delta s = 0$, $\Delta m_s = 0$

which means that there is NO change in the spin.

Now we concentrate on the terms

$$\langle n', l', m_{l'} | \vec{r} | n, l, m_l \rangle \propto \langle n', l', m_{l'} | \xi,\ \eta,\ z | n, l, m_l \rangle$$

Optical Transitions. Selection Rules

which amounts to the following integral

(Remember that: $\phi_{nlm_l}(r,\theta,\phi) = R_{n,l}(r)P_l^{m_l}(\cos\theta)\Phi_{m_l}(\phi)$)

$$\int_0^\infty dr\, R_{n',l'}R_{n,l}r^3 \int_0^\pi d\theta\{\sin^2\theta, \sin^2\theta, \sin\theta\cos\theta\}P_{l'}^{m_{l'}}(\cos\theta)P_l^{m_l}(\cos\theta) \times$$
$$\int_0^{2\pi} \left\{e^{[i(m_l-m_{l'}+1)\phi]}, e^{[i(m_l-m_{l'}-1)\phi]}, e^{[i(m_l-m_{l'})\phi]}\right\} d\phi$$

This last integral is NOT zero if

$$(m_l - m_{l'} + 1) = 0 \longrightarrow \Delta m_l = -1$$

$$(m_l - m_{l'} - 1) = 0 \longrightarrow \Delta m_l = 1$$

$$(m_l - m_{l'}) = 0 \longrightarrow \Delta m_l = 0$$

i.e. $\Delta m_l = 0, \pm 1$

Optical Transitions. Selection Rules

- In the case $\Delta m_l = 0$, we must also have

$$\int_0^{\pi} d\theta \ \sin\theta \ \cos\theta \ P_{l'}^{m_{l'}=m_l}(\cos\theta) P_l^{m_l}(\cos\theta)$$

is NOT zero. This will imply that $(l' - l) = \pm 1$, i.e. $\Delta l = \pm 1$.

- In the case $\Delta m_l = \pm 1$, we must also have

$$\int_0^{\pi} d\theta \ \sin^2\theta \ P_{l'}^{m_{l'}}(\cos\theta) P_l^{m_l}(\cos\theta)$$

is NOT zero. This will also imply that $(l' - l) = \pm 1$, i.e. $\Delta l = \pm 1$.

Both previous results are a consequence of the orthogonality of Legendre polinomials

- The integrals over *r* are always $\neq 0$, so there are NO selection rules for *n*.

Optical Transitions. Selection Rules

So, we finally obtain the selection rules.

$$\Delta s = 0$$

$$\Delta m_s = 0$$

$$\Delta m_l = 0, \pm 1$$

$$\Delta l = \pm 1$$

Optical Transitions. Selection Rules

The optical line spectrum of any LS coupling atom can be constructed from its energy level diagram by considering those transitions which follow some selection rules.

These rules refer to changes in the individual quantum numbers of the electron (as characterized within the Hartree approximation) as well as changes in the quantum number of the LS coupling.

- Transitions can occur only between configurations which differ in the *n* and *l* quantum numbers of a single electron. (from one subshell to another subshell)
- Transitions can occur when the change of subshell implies that the change in the individual l_i quantum number of the electron is such that

$$\Delta l_i = \pm 1$$

- Transitions can occur only between states in these configurations for which the changes in the s', l' and j' quantum numbers of the LS coupling satisfy

$\Delta s' = 0$ $\quad$ $\Delta l' = \pm 1$ $\quad$ $\Delta j' = 0, \pm 1$ $\quad$ (excepting $j' = 0$ to $j' = 0$)